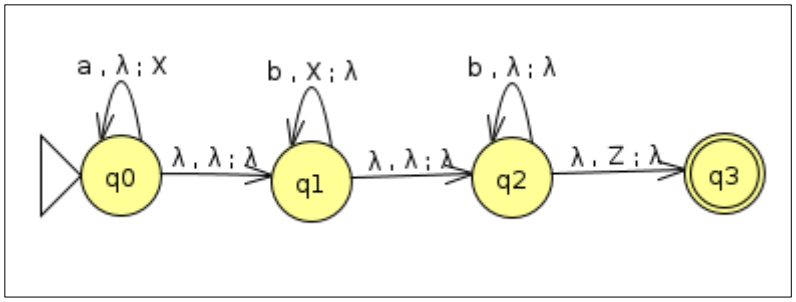
UNIVERSIDADE DO VALE DO ITAJAÍ
CURSO DE CIÊNCIA DA COMPUTAÇÃO
DISCIPLINA DE LINGUAGENS FORMAIS E AUTÔMATOS
PROFESSOR: ALEX LUCIANO ROESLER RESE, MSc.

Lista de Exercícios 12

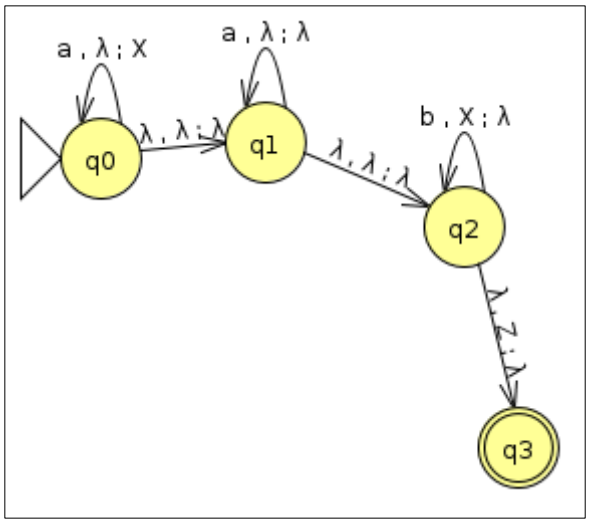
1. Crie Autômatos de Pilha e as Gramáticas Livres de Contexto para as linguagens a seguir:

   a) L4 = $\{a^x b^y \mid x <= y\}$



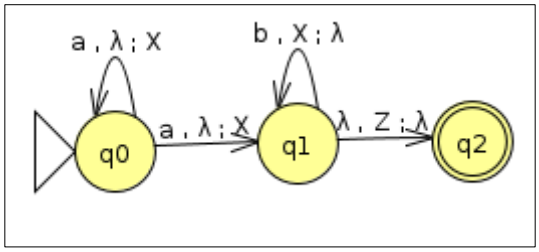
S → aAb | <epsilon>
A → S|Sb

   b) L5 = $\{a^x b^y \mid x >= y\}$



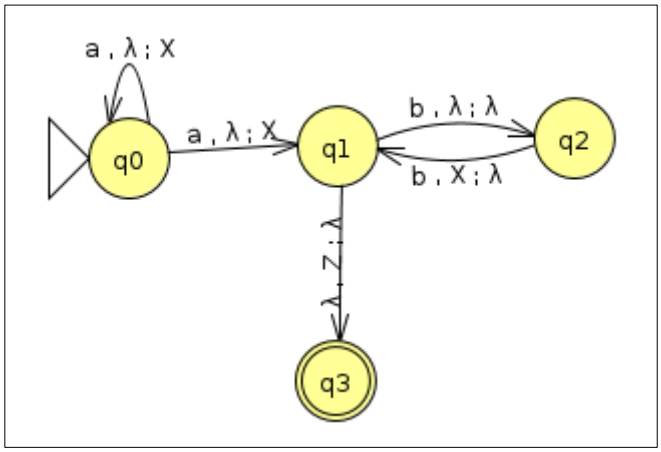
S → aAb | <epsilon>
A → S|aS

   c) L6 = $\{a^x b^x \mid x >= 1\} \cup \{a^x b^{2x} \mid x >= 1\}$



S → aAb

A → S | <epsilon>



S → aAbb

A → S | <epsilon>

2. Forneça a Forma Normal Chomsky para as GLCs abaixo:

a) S → aSa | bSb | a | b | aa | bb

S → CA | DB | a | b | aa | bb
A → SC
B → SD
C → a
D → b

b) S → bA | aB,
A → bAA | aS | a
B → aBB | bS | b

S → GA | EC
A → GD | ES | a
B → EF | GS | b
C → BH
D → AA
E → a
F → BB
G → b
H → ,

c) $S \rightarrow Aba$

$A \rightarrow aab$

$B \rightarrow AC$

$S \rightarrow AD$

$A \rightarrow CE$

$B \rightarrow b$

$C \rightarrow a$

$D \rightarrow BC$

$E \rightarrow CB$

d) $S \rightarrow 0A0 \mid 1B1 \mid BB$

$A \rightarrow C$

$B \rightarrow S \mid A$

$C \rightarrow S \mid$ <epsilon>

$S \rightarrow CE \mid DF \mid BB \mid CC \mid DD$

$A \rightarrow CE \mid DF \mid BB \mid CC \mid DD$

$B \rightarrow CE \mid DF \mid BB \mid CC \mid DD$

$C \rightarrow 0$

$D \rightarrow 1$

$E \rightarrow AC$

$F \rightarrow BD$

e) $S \rightarrow aAa \mid bBb \mid$ <epsilon>

$A \rightarrow C \mid a$

$B \rightarrow C \mid b$

$C \rightarrow CDE \mid$ <epsilon>

$D \rightarrow A \mid B \mid ab$

$S \rightarrow AC \mid BD \mid AA \mid BB$

$A \rightarrow a$

$B \rightarrow b$

$C \rightarrow AA$

$D \rightarrow BB$